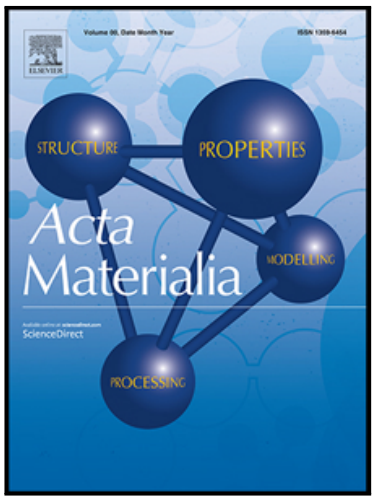
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Anisotropic deformation behavior of LPBF fabricated $Co_{35}Cr_{25}Mn_{15}Ni_{15}Fe_{10}$ with low stacking fault energy studied by in situ neutron diffraction

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Abstract

Stress-induced martensitic transformation (SIMT, FCC→HCP) is a crucial deformation mechanism that enhances ductility in alloys with low stacking fault energy (SFE), particularly in high-entropy alloys (HEAs). In this study, a low-SFE $Co_{35}Cr_{25}Mn_{15}Ni_{15}Fe_{10}$ HEA was fabricated via laser powder bed fusion (LPBF) to systematically investigated coupled effects of crystallographic orientation and grain size on SIMT during tensile deformation, using in situ neutron diffraction. <100>-oriented grains (D0) are found to facilitate dislocation dissociation through a higher accumulation rate of SFs during plastic deformation. In contrast, <111>-oriented grains (D45) exhibit an enhanced dislocation multiplication rate, leading to increased flow stress. However, D90, <110>-oriented grains, reveals grain size plays a more critical role in tailoring deformation mechanism than special orientation for low SFE alloy. D90 exhibits the highest phase fraction of martensitic after deformation attributed to the decreased apparent SFE by larger grain size. Anisotropic deformation mechanisms result in pronounced variations in mechanical properties: the highest ultimate tensile strength of 788 MPa is attributed to accelerated dislocation multiplication, while the SIMT activity contributes to extensive uniform plastic deformation, yielding a total elongation of 64.8%. These findings provide mechanistic insights into the deformation behavior of low-SFE alloys and have broad implications for microstructural design—particularly through controlled orientation and grain size—toward achieving exceptional mechanical properties.

Keywords: High entropy alloy; Stacking fault energy; Martensitic transformation; Laser Powder Bed Fusion; Deformation mechanism

1 Introduction

CoCrFeNiMn-based alloy is a series of widely researched face-centered cubic (FCC) single-phase high-entropy alloys (HEAs), which has a broad application prospect in aerospace and marine fields attributed to excellent thermal and mechanical properties especially as superior fracture resistence at cryogenic temperatures [1–5]. For the alloy, dislocation slip plays as the main plastic deformation mechanism at room temperature [6,7]. As temperature decreasing, the reduced stacking fault energy (SFE) induces deformation twinning, meanwhile lower SFE tends to stablize the HCP phase. Thus, mechanical behavior is intrinsically governed by SFE for FCC HEA. As the SFE decreases from high to critically low levels in the CoCrFeNiMn high-entropy alloy system [8–10], the dominant plastic deformation mechanisms evolve sequentially from dislocation slip [11] to deformation twinning, and ultimately to martensitic transformation (FCC → HCP) [8,12,13].

In $Co_{20}Cr_{20}Fe_{20}Mn_{20}Ni_{20}$ with a SFE of -72 mJ/m^2 at 0 K based on density functional theory (DFT) method, dislocation glide and twinning have been identified as the dominant deformation mechanisms [14–16]. Elevating the ratio of Co and Cr is an effective method to lower the SFE [8–10]. In our previous study, $Co_{35}Cr_{25}Mn_{15}Ni_{15}Fe_{10}$ with low level SFE (-87 mJ/m^2 at 0 K based on DFT method) was designed and its deformation process was primarily driven by FCC → HCP phase transformation [8]. Rolled-annealed samples perform superior combination with strength and ductility, which is attributed to high ratio of stress induced martensitic transformation (SIMT) [17]. These two phases follow an orientation relationship described as: $\{111\}_{FCC}$ // $(0001)_{HCP}$, $<110>_{FCC}$ // $[11\bar{2}0]_{HCP}$ [10]. Naturally whether the crystallic orientation influences the SIMT activity also is a research point, which remains a contriversial issue. Some researchers argue that the <100> orientation, with its high Schmid factor, activates multiple slip systems and promotes local stress concentration, thereby favoring twinning and transformation[18,19]. Others holds that grains with high Taylor factors (e.g., <111>, <110>) activate multiple slip more easily, whereas the <100> orientation, dominated by a single slip system, lacks the multi-slip necessary for twinning/transformation[20,21].

Laser Powder Bed Fusion (LPBF) has recently been applied to fabricate HEAs, owing to its layer-by-layer construction capability, which enables the production of complex metallic structures [20,21]. The fixed thermal flow direction during every layer contributes to distinct microstructure, which is similar to samples fabricated by directional solidification. Special thermal flow patterns

realize the controlling of grains orientation [22]. By exploiting this characteristic, compositions with relatively high SFE, such as LPBFed CoCrFeMnNi [23] and $(CoCrFeMnNi)_{99}C_1$ [24], were processed to obtain samples exhibiting pronounced preferred orientations. In particular, a {001} texture with the ⟨100⟩ direction parallel to the building direction (BD) was achieved. On this basis, it was demonstrated that the competition between dislocation slip and deformation twinning leads to markedly different deformation behaviors depending on the crystallographic orientation. As for low SFE alloys, SIMT acting as primary mechanism also performed the anisotropic behavior. While as mentioned above, this issue remains a matter of debate.

Another kind of representative feature in LPBFed HEA is dendritic and/or columnar grains [25–28]. Obviously distinct microstructures further influence the mechanical performance. For instance, samples loaded along the scanning direction exhibited a higher yield strength (741 MPa) by profuse evolution of geometrically necessary dislocation (GND) at the denser boundaries compared to those loaded perpendicularly (694 MPa), indicating the significant role of grain morphology in governing mechanical response [24]. In this work, a series of samples with preferred orientations were fabricated by exploiting the intrinsic characteristics of LPBF. The degree of SIMT activation varies markedly among these samples and this orientation dependent deformation behavior was elucidated by in-situ neutron diffraction and transmission electron microscopy.

2 Materials and methods

2.1 Manufacturing process

Original blocks were firstly mixed by pure elemental blocks of Co, Cr, Mn, Ni, and Fe according to the nominal atomic ratio, each with a purity of 99.95 wt. %. And these blocks were melted to form a homogeneous liquid $Co_{35}Cr_{25}Mn_{15}Ni_{15}Fe_{10}$ alloy. The liquid alloy was then gas atomized to produce the final powder, as shown in Fig. 1a. The oxygen content of the as atomized powder was measured to be below 800 ppm using an oxygen analyzer (LECO). Fig.1b was the schematic diagram of LPBF process and stainless-steel plate acted as the substrate. The $Co_{35}Cr_{25}Mn_{15}Ni_{15}Fe_{10}$ bulk was manufactured with a maximum laser intensity of 300 W and scanning speed of 800 mm/s. Layer thickness and hatch distance were fixed at $t = 40$ μm and $h = 90$ μm. Each layer was scanned in a zigzag-type pattern and the scanning direction was rotated 67° between (N)th and (N+1)th layers. After the LPBF process, the sample with a dimension of 35

× 35 × 87 mm^3 was deposited in the argon atmosphere. Following that, annealing was carried out at 693 K for 6 h using an electrical resistance furnace. And the element composition of LPBFed samples is provided by energy dispersive spectrometer (EDS) in Table 1.

Table 1 The EDS-obtained elemental compositions of the investigated HEAs.

Element/at. %	Co	Cr	Mn	Ni	Fe
As-deposited	33.6±0.1	26.1±0.1	13.6±0.2	14.5±0.1	12.0±0.2

2.2 Mechanical properties tests

To investigate the anisotropic deformation behavior and mechanical properties, tensile specimens were sectioned along three specific orientations—namely, 0°, 45°, and 90° relative to the x-axis (defined in Fig. 1c)—from a 2-mm-thick slice cut from the xoz plane. These specimens are designated as D0, D45, and D90, respectively. Tensile tests were performed with a screw-driven tensile testing machine (SHIMADZU AGS-X 50 KN) with an engineering strain rate of $10^{-3}s^{-1}$ at room temperature. Full range extensometer (SHIMADZU Non-Contact Digital Video Extensometer) was used for strain measurement.

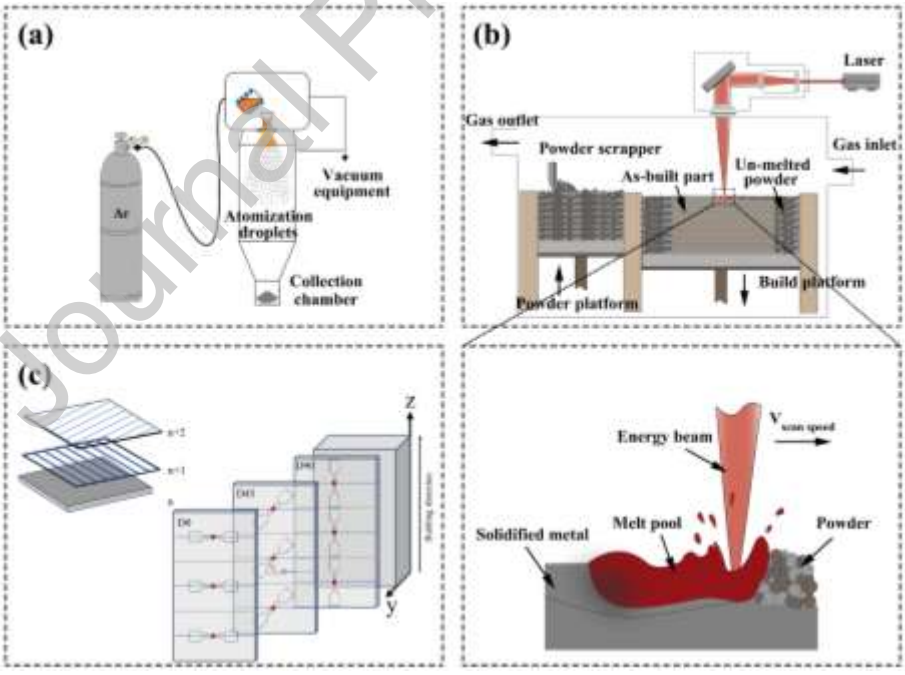


Fig.1 **The schematic diagram of construction and cutting strategy of dog-bones specimens.** Powder atomization(a), laser powder bed fusion (LPBF) (b), cutting strategy (c).

2.3 Microstructural characterization

XRD patterns were collected on a Rigaku Rapid IIR micro-diffractometer (Rigaku, Japan)

using Cu Kα radiation (λ = 0.154 nm) at 40 kV and 250 mA. The beam collimator diameter was 0.1 mm and the diffraction patterns were recorded over a 2θ range of 30–80° at a scanning speed of 2°/min. The Jade 6.0 software was used to analyze the XRD data [29]. Specimens as-built and after tensile fracture were then embedded in an epoxy resin for the gauge section microscopic studies. Samples were mechanically ground using 600, 1200, and 2000 grit SiC abrasive papers, followed by vibratory polishing. The vibratory polishing was performed first with a 0.5 μm alumina suspension for 12 hours, and then with a 0.02 μm colloidal silica suspension for another 12 hours. Subsequent characterization was conducted using a JEOL IT800 scanning electron microscope equipped with an Electron backscattered diffraction (EBSD) detector and an EDS. OIM software was employed for EBSD analysis [30]. Disks of phi 3 μm cut from these samples' gauge were mechanically ground to ~40 μm thickness, followed by Argon ion milling (PIPS II 695, Gatan) initially at 5 keV, and then a final step at 2 keV to reduce surface damage. And then transmission electron microscope (TEM, Thermo Scientific Themis Z) observation was operated at an accelerating voltage of 300 kV.

In-situ neutron diffraction measurements during tensile deformation of the HEAs were performed using the time-of-flight diffractometer at beamline BL19 "TAKUMI" in the Materials and Life Science Experimental Facility (MLF) at J-PARC (Japan Proton Accelerator Research Complex). A neutron wavelength range of 0.75–4.5 Å was selected, providing a neutron flux of approximately 2–3 × 10^7 $n \cdot s^{-1} \cdot cm^{-2}$ at the sample position. The loading direction was inclined at +45° relative to the incident neutron beam. A rectangular incident beam with dimensions of 5 mm (width) × 5 mm (height) was employed. Two detector banks equipped with 5 mm-wide radial collimators were positioned at −90° and +90° to the incident beam, corresponding to the axial and transverse orientations of the specimen, respectively [31]. Each detector covered a solid angle of ±15° in both horizontal and vertical directions. The diffraction data spanned a d-spacing range of 0.05–0.30 nm with a resolution of $\Delta d/d \approx 0.3\%$. {hkl}-specific d-spacings and inverse pole figures were extracted from the collected profiles by Rietveld refinement (Z-Rietveld code) [32,33]. Stacking fault probability (SFP) was evaluated by tracking the peak shifts among crystallographically equivalent reflections with the scattering vector aligned parallel to the loading axis [34,35]. Finally, the dislocation density was determined using the convolutional multiple whole profile (CMWP) analysis method.

3 Results and discussion

3.1 Microstructural characteristics and properties

3.1.1 Microstructural characteristics

Due to the specific cutting strategy employed, only the morphology of D90 on x-z plane is presented in Fig.2a and Fig.S1a. The microstructure is free of visible cracks or defects. Samples exhibited a columnar grain structure aligned with the building direction, which is the characteristic of localized solidification under high thermal gradient and rapid solidification rate [36,37]. As shown in Fig.2b, only FCC phase peaks were detected in the neutron diffraction pattern. And the results from XRD patterns and EBSD also show a single FCC phase in as-built samples (Fig.S1b and f-h, respectively). Fig.2c-e presents inverse pole figures of grains taken along the loading direction D90 (c), D45 (d), and D0 (e), respectively. These show different planes perpendicular to the loading direction in each case. The D90 specimen (Fig.2c) exhibits a strong preferential growth orientation along the <110> and <100> crystallographic direction evidenced its by inverse pole figure, similar to other cubic materials fabricated by additive manufacturing [23,38–41]. This phenomenon results from the epitaxial growth mechanism of grains across the melt-pools for cubic materials [23,41]. Compared to D90, D0 and D45 samples exhibit significant differences not only in their columnar crystals grain arrangements but also in their dominant crystal orientation. The preferred grain orientation is predominantly <100> and <311> in D0 (Fig.2e) and <111> in D45 (Fig.2d), respectively.

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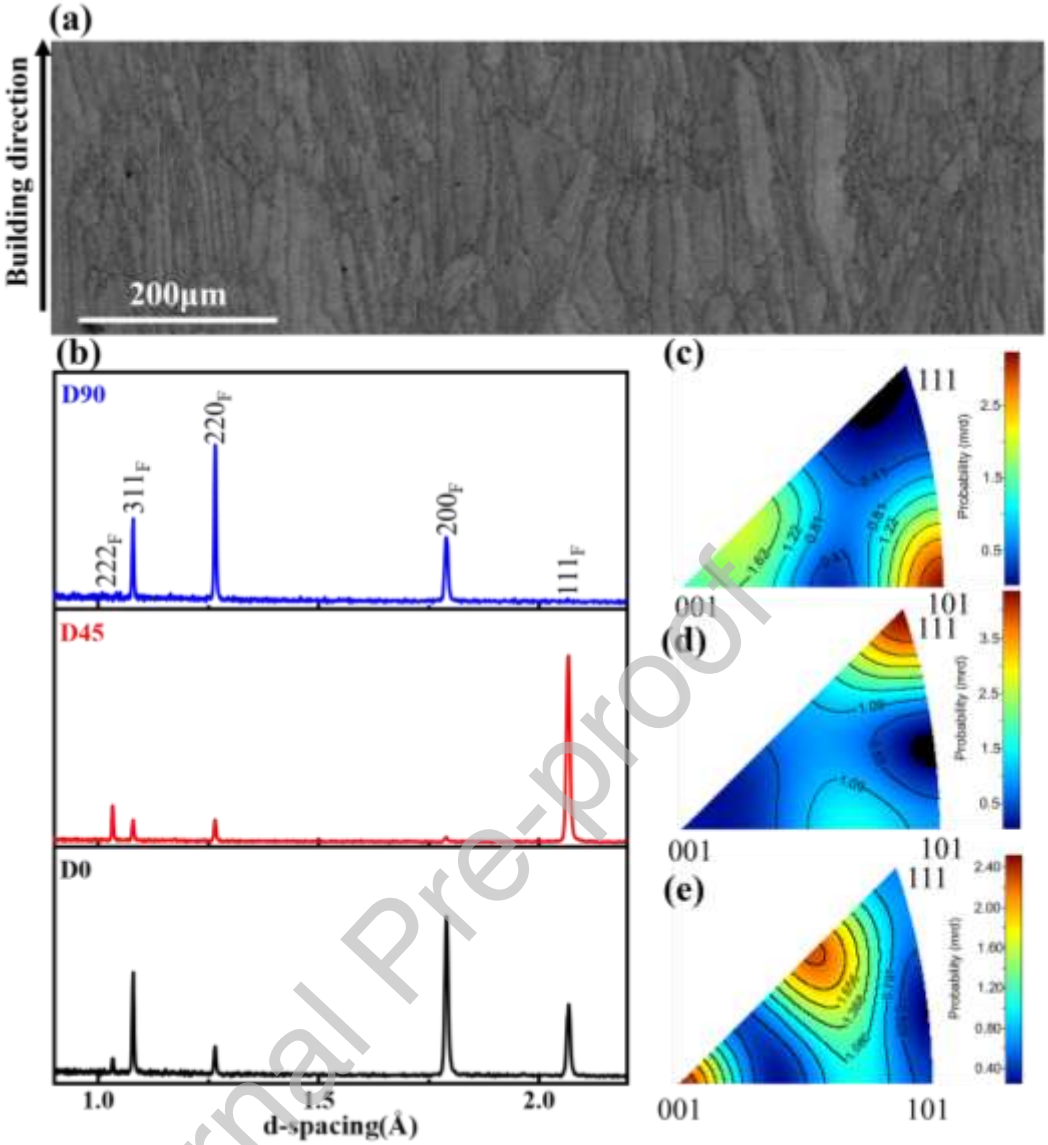


Fig.2 **Morphology and crystal structure of the undeformed as-deposited HEA** (a) SEM image of D90 on x-z plane. (b) Neutron diffraction patterns of all samples. Inverse pole figures show the texture of the grains along the loading direction before deformation D90 (c), D45 (d) and D0 (e).

For LPBFed bulks, the special microstructure stems from the cyclic rapid thermal loadings. In high SFE alloy, large thermal stress induces multi-tangled dislocation cells. As SFE decreasing, twins and HCP phases also can be induced in the matrix [15]. And then the nano-scale microstructure of as-built alloy was characterized using TEM as depicted in Fig.3. In Fig.3a, rich dislocations were tangled to form a typical cellular sub-structure with a width of approximately 400 nm and a thickness of around 80 nm. The selected area electron diffraction (SAED) in the insert of Fig.3a confirms the presence of an FCC phase, indicated by a single set of diffraction spots. Within these columnar dislocation cells, it is worth noting that a distinctive network-like feature was observed, marked by white arrows in Fig.3a and b. To further clarify the nature of this network, the HRTEM (high

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resolution TEM) was performed on the region enclosed by the white rectangle in Fig.3b. The HRTEM image (Fig.3c) reveals that these parallel and intersecting lines correspond to stacking faults (SFs) and Lomer–Cottrell (L–C) locks, marked by blue arrows. In addition to the SFs, several thin ribbons were observed adjacent to the network (Fig.3d), which were identified as nano HCP phase via SAED (Fig.3d) and HRTEM (Fig.3e).

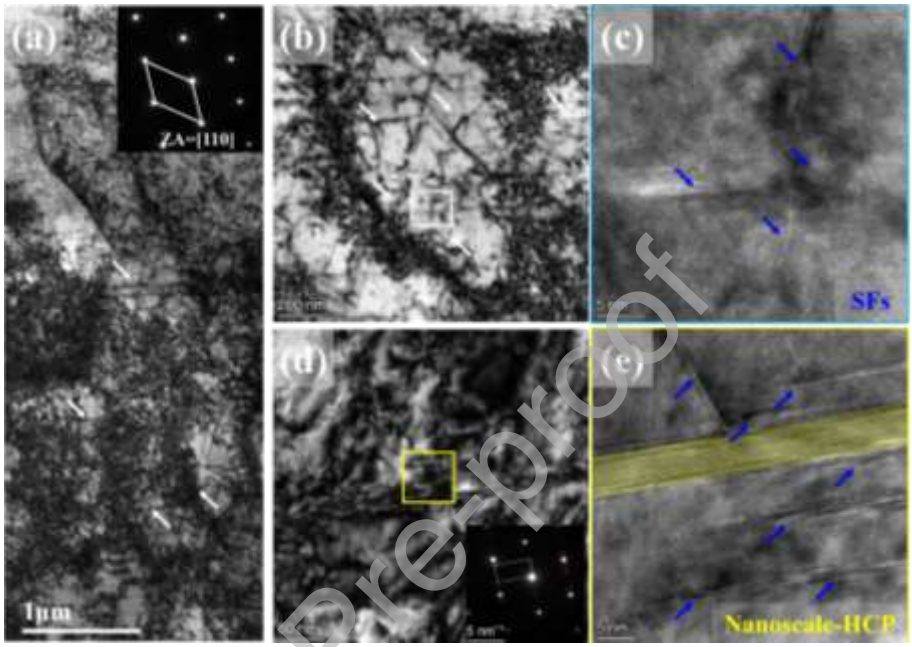


Fig.3 **Abundant dislocation cells, stacking fault networks and nanotwins induced by thermal stress and low stacking fault energy.** Bright field (BF) (Insert: SAED patterns) of as-built $Co_{35}Cr_{25}Mn_{15}Ni_{15}Fe_{10}$ (a, b, d), the HRTEM image of the area denoted by white rectangle in (b) is shown in (c). The HRTEM image of the area denoted by yellow rectangle in (d) is shown in (e).

The formation of SF networks and HCP phase could be analyzed from two primary factors: SFE and thermal stress. For $Co_{35}Cr_{25}Mn_{15}Ni_{15}Fe_{10}$, its intrinsically low SFE [8] facilitates the dissociation of perfect dislocations into Shockley partial dislocations [42], thereby enhancing SF formation. In other low SFE alloys, similar SF network structure was also observed in deformed samples activated by the mechanical stress [43,44]. The LPBF process is well-known for inducing steep thermal gradients and high solidification rates during localized melting and solidification, which leads to significant thermal stress accumulation and the formation of SF networks [45]. In the meanwhile, these formed locks enhance strain hardening and strength by acting as strong barriers to dislocation motion [43].

The HCP phase was frequently observed in proximity to stacking faults, forming an angle of approximately 70° with them. And they lie on two $\{111\}_{FCC}$ planes identified by HRTEM

observations. The orientation relationship between the FCC and HCP phases, as determined by SAED, is consistent with the classical correspondence: $<110>_{FCC}//<11\bar{2}0>_{HCP}$ and $\{111\}_{FCC}//(0001)_{HCP}$, as previous researches about HEAs discussed [10]. The atomic arrangement along the zone axis ZA=<110> in the FCC phase follows an ABCABC stacking sequence. And along the <1120> direction in the HCP phase, it exhibits an ACACAC stacking sequence. It is widely accepted that the HCP phase can form through the coordinated glide of Shockley partial dislocations on every second $\{111\}_{FCC}$ plane [10]. Naturally, low SFE is beneficial for HCP forming during building process. However, these nano HCP phase often remains undetectable by conventional XRD and EBSD due to its limited volume fraction and nanoscale dimensions.

3.1.2 Anisotropy in tensile properties

Tensile tests were conducted at 293 K to clarify the anisotropy of mechanical properties. Engineering stress-strain and true stress-strain curves are depicted in Fig.4(a and b, respectively), demonstrating clear sensitivity to the loading direction. Detailed mechanical properties are summarized in Table 2. The strain hardening rate (SHR) curves exhibit a pronounced initial increase in all samples during early deformation. This early-stage hardening behavior is notably more significant than that reported in other LPBFed CoCrFeMnNi HEAs [14,18,24,38]. The strain hardening coefficients ($\gamma = \Delta\sigma/\Delta\varepsilon$), calculated from the engineering stress–strain curves [46], are estimated as 1.07 GPa for D0, 0.79 GPa for D45, and 0.37 GPa for D90. A strong directional dependence is observed in the work hardening behavior.

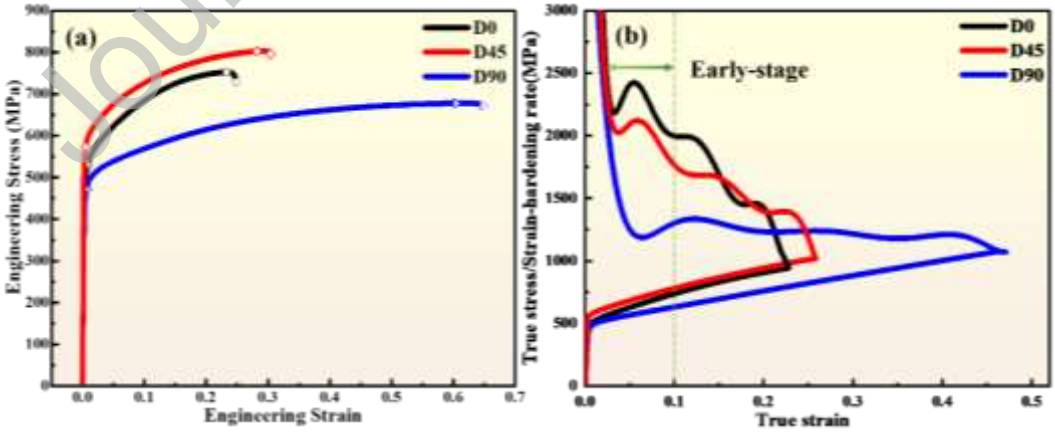


Fig.4. **The mechanical properties of HEAs under different loading directions.** (a) The engineering stress-strain curved. (b) The corresponding true stress-strain curves and strain hardening rate curves.

The yield strength (YS), ultimate tensile strength (UTS), and elongation (EL) values are summarized in Table 2. Among these tested samples, D45 exhibits the highest strength, with an YS

of 557 MPa and a UTS of 788 MPa, while still retaining good plasticity (total elongation: 32.8%). In contrast, D90 shows the highest total elongation, reaching 64.8%, though with lower strength values of 467 MPa (YS) and 678 MPa (UTS). D0 demonstrates intermediate strength properties, with an YS of 478 MPa and a UTS of 751 MPa, both higher than those of D90.

Table 2 The mechanical properties of HEAs under different loading directions.

Sample	Yield strength/MPa	Elongation/%	Ultimate tensile strength/MPa
D0	478±26.8	28.0±1.95	751±13.7
D45	557±19.3	32.8±0.97	788±13.5
D90	467±7.0	64.8±4.12	678±2.9

The most significant difference in YS is observed between samples D45 and D90, with a margin of approximately 100 MPa. This discrepancy primarily arises from variations in solid-solution strengthening, dislocation strengthening, and grain boundary strengthening. For these samples under different loading directions, the contributions of solid-solution and initial dislocation strengthening are considered comparable. The most distinguishing factor lies in the grain arrangement relative to the loading direction, which influences both the number of grain boundaries and the crystallographic orientation. The D45 sample, which displayed the highest YS, possesses a higher density of grain boundaries and a hard orientation with lower Schmid factor for slipping, thereby enhancing the probability of dislocation blocking [18]. Concurrently, it is well established that an alignment of the loading direction with the <110> and <100> crystal orientation results in reduced yield strength [15]. In D9, the lowest YS originates from the soft orientation (<100> and <110>) and the low grain boundary density.

3.2 In-situ neutron diffraction investigation

In-situ tensile tests were conducted using neutron diffraction (ND) to investigate the plastic-deformation behavior under three different loading directions. Fig. 5 shows the stacked ND profiles for D0 (Fig. 5a), D45 (Fig. 5b), and D90 (Fig. 5c) at various true strain levels, which were arranged by the increasing strain. The specific strain for each curve was also labeled above the curve. And undeformed samples exhibited different preferred orientations, consistent with the experimental design. In D0, $\{200\}_F$ and $\{311\}_F$ planes were the main orientations. For D45, $\{111\}_F$ orientation

was predominant, whereas in D90, $\{111\}_F$ peaks were hard to be detected; instead, $\{220\}_F$ and $\{200\}_F$ orientations dominated. With increasing strain, diffraction peak broadening occurred in all three samples, and the formation of HCP phase was observed after tensile deformation, confirming the occurrence of SIMT.

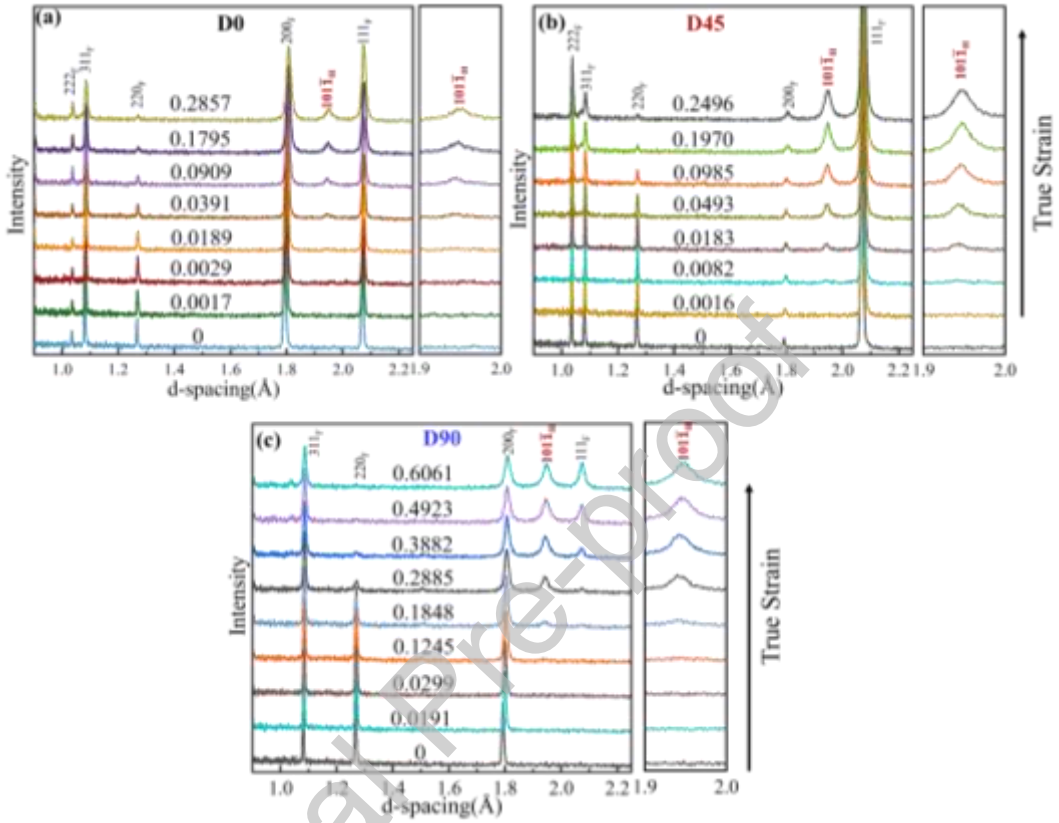


Fig. 5 **Crystal structure evolution during tensile loading for samples revealed by in-situ neutron diffraction (ND):** D0 (a), D45 (b), and D90 (c).

Fig. 6 shows the lattice strain of D0 (a), D45 (b), D90 (c) samples, analyzed from the ND profiles, using [47]

$$\varepsilon_{hkl} = \frac{d_{hkl} - d^0_{hkl}}{d^0_{hkl}} \quad (1)$$

where ε_{hkl} is the lattice strain of {hkl} planes, d_{hkl} is the lattice spacing of {hkl} planes at a given strain, d^0_{hkl} is the lattice spacing of {hkl} planes before the tensile test. It can be observed that the lattice strains of all the $\{111\}_F$, $\{200\}_F$, $\{220\}_F$, $\{311\}_F$, $\{222\}_F$, and $\{400\}_F$ planes increase with the increase in true strains. The lattice strains follow the sequence of $\varepsilon_{\{220\}}<\varepsilon_{\{111\}}<\varepsilon_{\{222\}}<\varepsilon_{\{311\}}<\varepsilon_{\{200\}}$ in D0 and D45 samples. Whilst in D90 sample, the origin state of $\{111\}_F$ was too weak to calculate compared to the other peaks and the lattice strains follow the

sequence of $\varepsilon_{\{111\}}<\varepsilon_{\{220\}}<\varepsilon_{\{311\}}<\varepsilon_{\{200\}}$. These results indicate that the <200> orientation is softer than the <220> and <111> orientations. In addition, the $\varepsilon_{\{hkl\}}$ of the D0 is larger than those of the D45 and D90 at an equivalent strain. In theory, the {111} and {222} planes are crystallographically equivalent (parallel planes) in an FCC lattice, and the lattice strains measured from these two reflections should be identical [48]. However, the presence of stacking faults disrupts the lattice periodicity, leading to a measurable difference in the lattice strains derived from the {111} and {222} reflections.

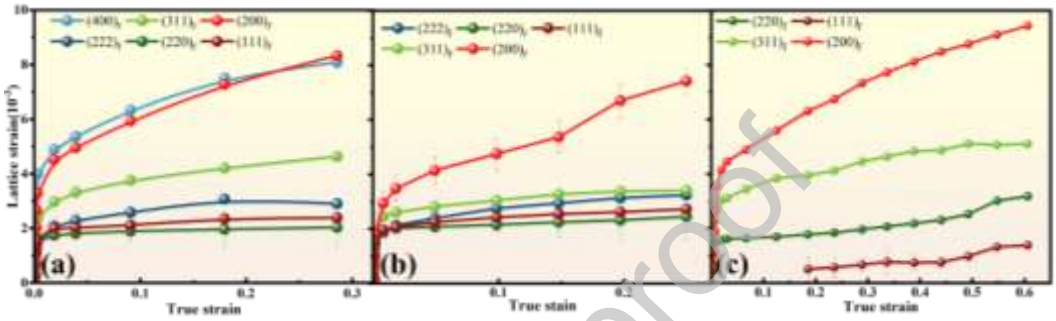


Fig. 6 **Lattice strain at various tensile strains:** D0 (a), D45 (b), and D90(c) HEAs.

Based on the lattice strain, we acquired the stacking fault probability (P_{sf}), as follows [49, 50]:

$$P_{sf} = \frac{32\pi}{3\sqrt{3}}\left(\frac{d_{\{222\}} - d^0_{\{222\}}}{d^0_{\{222\}}} - \frac{d_{\{111\}} - d^0_{\{111\}}}{d^0_{\{111\}}}\right) = \frac{32\pi}{3\sqrt{3}}\left(\varepsilon_{\{222\}} - \varepsilon_{\{111\}}\right) \quad (2)$$

For the D90 sample, the $\{222\}_F$ and $\{111\}_F$ diffraction peaks could not be collected due to the strong crystallographic texture, and thus its P_{sf} is absent in Fig. 7a. As shown in the figure, P_{sf} increases with strain in both D0 and D45. While the P_{sf} value in D0 increases more rapidly. The maximum P_{sf} values reached are 15.3×10^{-3} for D0 (at a strain of 0.286) and 10.6×10^{-3} for D45 (at a strain of 0.250).

The dislocation density (ρ) of these HEAs during the deformation is presented in Fig. 7b. ρ increases more rapidly with the strain in D0 and D45 than in D90, which followed the trend: D45 > D0 > D90. At the fracture stage, ρ reached a nearly maximum value of 47.1×10^{14} m^{-2} in D45 at a strain of 0.250, 48.5×10^{14} m^{-2} in D0 at a strain of 0.286 and 49.6×10^{14} m^{-2} in D90 at a strain of 0.547, respectively. Since both samples possess the same SFE, the difference in P_{sf} and dislocation density evolution can be attributed to variations in the dominant crystallographic orientation and grain size aligned with the loading direction.

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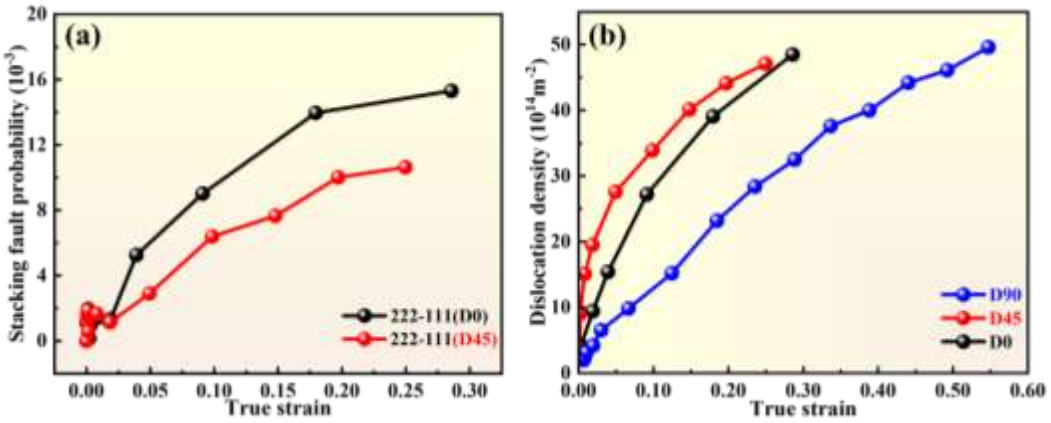


Fig. 7 **Stacking faults and dislocation density evolution during tensile loading.** (a)Stacking fault probability (P_{SF}) acquired from Eq. (2) acquired from the CMWP process of the D0 and D45 HEAs at various tensile strains; (b) Dislocation density of the D0, D45, and D90 HEAs in-situ tensile deformed to various strains.

The relative diffraction peak intensity, obtained by normalizing the real-time intensity (I) by its initial value (I_0), was calculated to track texture evolution and is plotted in Fig. 8. Generally, the evolution of diffraction peak intensity is attributed to grain reorientation during deformation. [47]. A pronounced decline in the relative intensity of the $(220)_F$ and $(311)_F$ peaks was observed upon macroscopic yielding, consistent with previous reports on FCC systems [51,52]. In contrast, the intensity evolution of the $(200)_F$, $(111)_F$, and $(222)_F$ peaks exhibited more complex behavior, suggesting the activation of additional deformation mechanisms.

Prior to a strain of 0.02, the I/I_0 ratio of the $(200)_F$ peak exhibited minor fluctuations, followed by a gradual decrease in all samples. The initial mild increase aligns with the development of deformation texture driven primarily by dislocation slip in FCC alloys [51,53]. Typically, the intensities of the $(111)_F$ and $(222)_F$ peaks increase monotonically with strain due to grain rotation [47,51,53]. Conversely, a decrease in $I/I_0(111)_F$ can be associated with the consumption of the FCC phase due to FCC→HCP transformation [47], as supported by the emergence of HCP peaks in Fig. 5. The subsequent gradual increase at higher strains suggests that dislocation multiplication and associated texture evolution became the dominant deformation mechanisms in D0. For D45, the <111>-oriented grains remain the dominant orientation. While in polycrystalline deformation, grain orientation is also affected by adjacent grains, and the resulting intergranular compatibility can lead to a decrease in the $I/I_0(111)_F$. In D90, a general reduction in all FCC peak intensities was observed, resulting from extensive phase transformation. The slight recovery of $I/I_0(111)_F$ at large strains ($\varepsilon \approx 0.1245$) is likely due to remaining FCC phase rotating.

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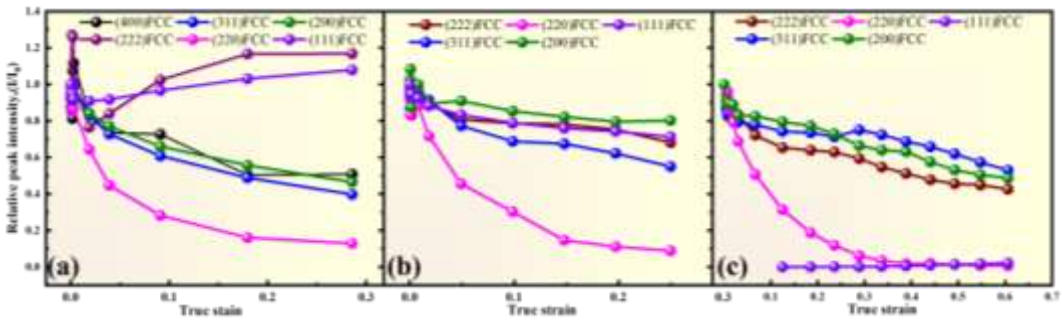


Fig. 8 **Normalized peak intensity (I/I_0) evolution during in-situ tensile deformation:** D0 (a), D45 (b), and D90 (c) samples.

The XRD results of the fractured samples are presented in Fig.9a. Characteristic diffraction peaks of the HCP phase were detected in all samples, with the highest relative intensity observed in D90. The HCP phase content, as quantified from EBSD phase maps (Fig.9 b1-d1), was determined to be 4.0% in D0, 3.6% in D45 and 33.3% in D90. These results indicate a strong dependence of phase transformation capacity on the loading direction: transformation is enhanced when the loading direction aligns more closely with building direction.

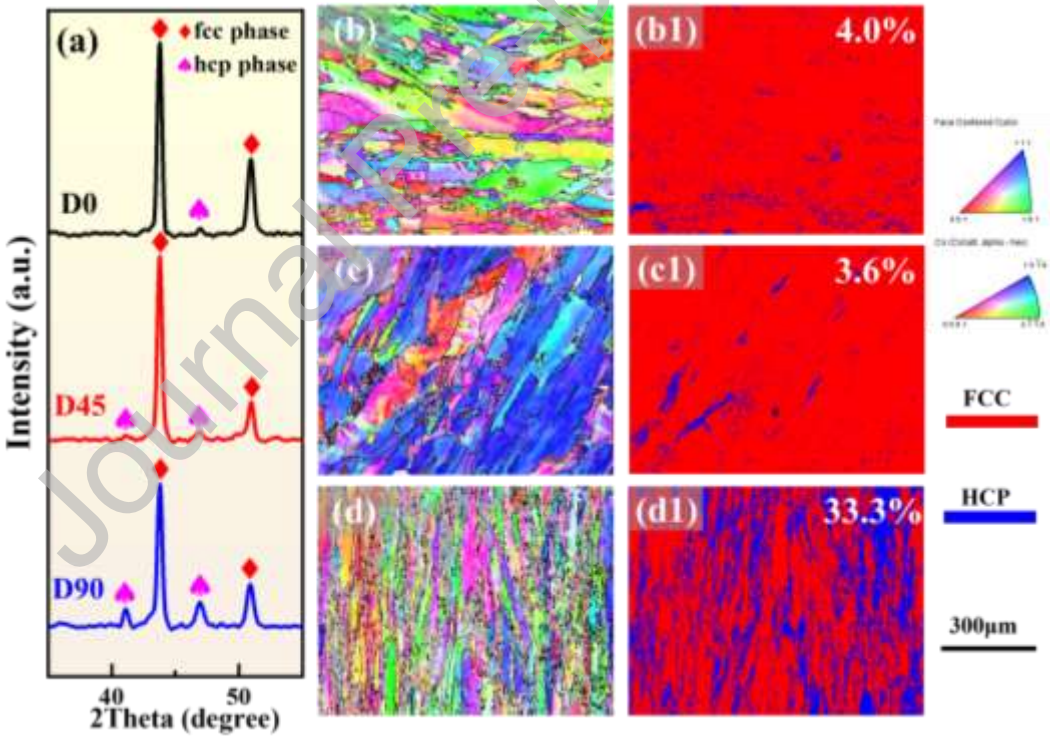


Fig.9. **The HCP content of fractured specimens is strongly dependent on the loading direction**. (a) XRD patterns of fractured samples. (b-d) IPF, (b1-d1) corresponding phase maps for D0, D45 and D90, respectively.

3.3 Orientation-dependence plastic deformation mechanism

3.3.1 Deformation Microstructure

The in-situ neutron diffraction results demonstrate that plastic deformation in these samples is

mediated by a complex interplay of dislocation multiplication, SF formation, and FCC→HCP phase transformation. TEM observations provide direct microstructural evidence for these deformation mechanisms.

Based on the SHR curves, samples D0 and D45 exhibit similar mechanical response. Therefore, detailed microstructural analysis was carried out only for D45 and D90 at 10% strain, after the first working hardening, as shown in Fig. 10 and Fig. 11, respectively. After 10% tensile strain, D45 contains abundant deformation bands with a high density of dislocations, marked by green arrows (Fig. 10a). The corresponding SAED pattern from this region reveals this region was a single FCC phase. And these pre-existing deformation features, cellular dislocation structures and SF networks, were annihilated during the initial tensile deformation. And no new dislocation cells were produced by tensile stress, which is attributed to the low SFE. The low SFE promotes planar slip of dissociated dislocations and strongly suppresses dynamic recovery processes—such as cross-slip and dislocation climb—that are necessary for dislocation cell formation [54].

Within these deformation bands, abundant SFs were also identified by streaks in diffraction patterns (the insert in Fig. 10a) and L–C locks were also identified via HRTEM (Fig. 10b). It can be inferred that such structures also contribute to work hardening in the present alloy. And this effect is especially pronounced when compared to LPBFed CoCrFeMnNi, which exhibit higher SFE and a general absence of such extensive SF networks after printed [38].

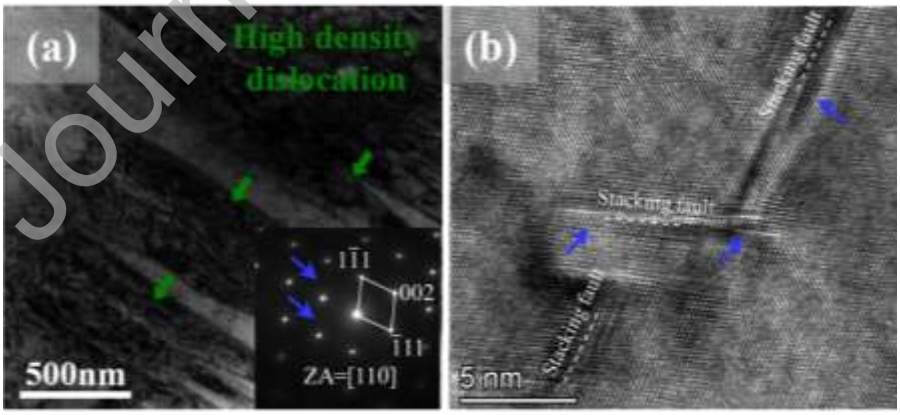


Fig. 10 **Profuse dislocations and SFs in D45 at 10% strain**. (a) BF image of deformed sample and the insert is the SAED pattern. (b) HRTEM images of SFs.

In addition to the deformation bands presented in Fig. 11a, a distinct ribbon-like morphology was observed in the D90 sample deformed to 10% tensile strain (Fig. 11b and d). SAED analysis of this region, denoted by red circle in Fig. 11b, confirmed the presence of twins evidenced by additional diffraction spots (Fig. 11c). HRTEM further verified the formation of HCP phase within

these ribbon features (Fig. 11e), unambiguously identified through FFT analysis (Fig. 11g and h). Moreover, abundant SFs were observed adjacent to these nanotwins, as shown in Fig. 11f.

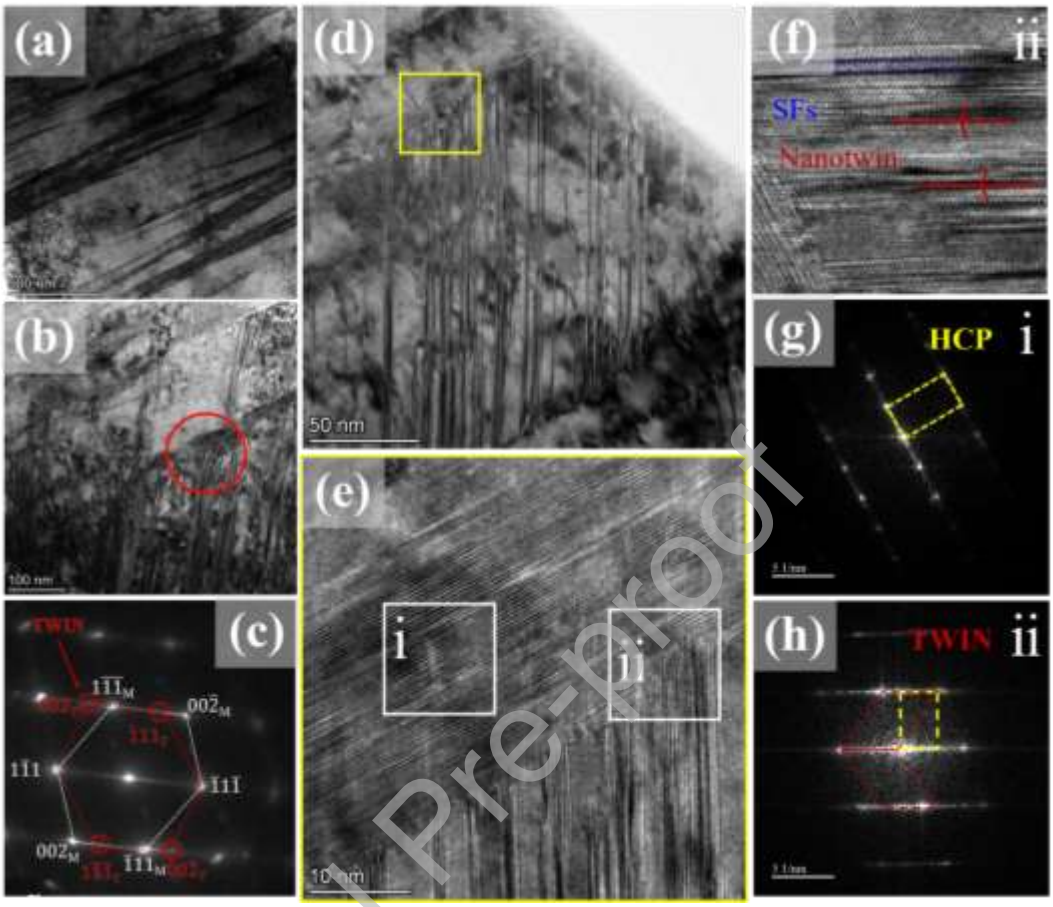


Fig. 11 **Abundant deformation products in D90 at 10% strain: HCP, nanotwins and SFs.** (a, b, d) BF images of deformed sample. (c) is the SAED of the area denoted by red circle in (b). (e) is the HRTEM of the area denoted by yellow rectangle in (d), (f) is the enlarge image of the area denoted by white rectangle (ii) and (g, h) are FFT results of areas denoted by white rectangles (i, ii, respectively).

Given the limited field of view inherent to TEM techniques, EBSD was additionally employed to investigate microstructural evolution at a larger scale. Phase maps derived from EBSD confirmed the activation of phase transformation in both D45 and D90 samples at 10% strain in Fig. 12. In both D0 and D45, the HCP phase was detected via neutron diffraction prior to 10% strain. For the D90 sample, although HCP diffraction peaks were not detected until a strain of 12.45%, TEM and EBSD observations confirm that SIMT had already been activated at approximately 10% strain. Although deformation products are composed by dislocation, SFs, TWINs and HCP, their ratio in fractured samples is different identified by neutron diffraction and EBSD, which point to different dominant deformation mechanism.

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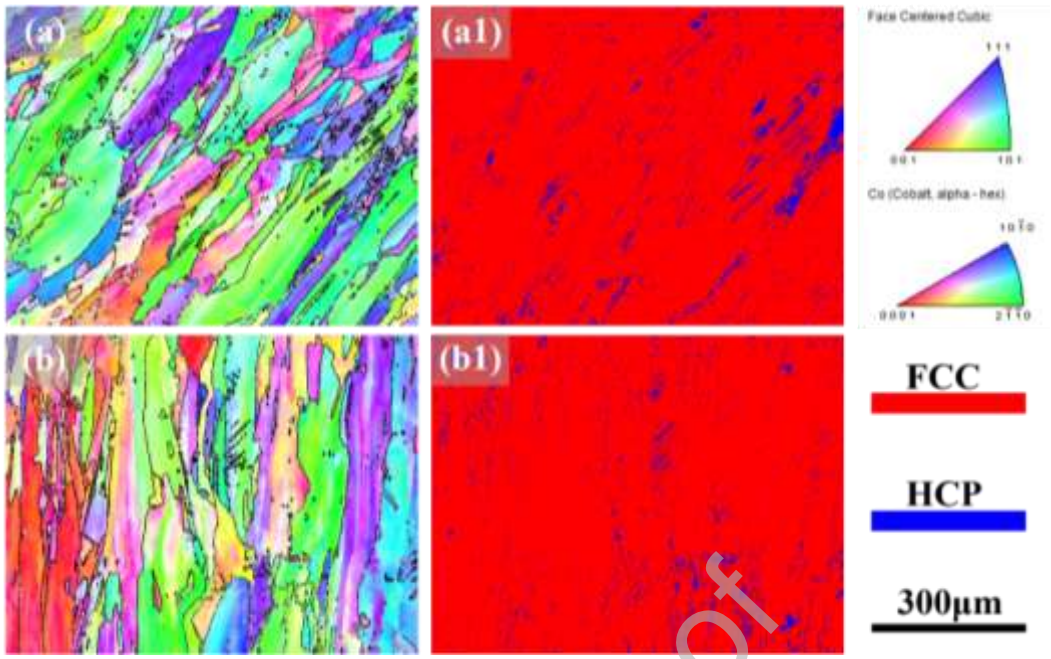


Fig. 12 **SIMT activated in both D45 and D90 with 10% strain.** (a-b) IPF, (a1-b1) corresponding phase maps for D45 and D90, respectively.

3.3.2 The origin of anisotropic mechanical responses

D45 performs the highest dislocation multiplication rate among them, indicating that dislocations tended to glide and multiplicate. This can be attributed to its dominant orientation, <111>. This promoted strong interactions among dislocations from multiple slip systems, resulting in the formation of trapping mobile dislocations barriers such as L-C locks. And high dislocation multiplication rate leads to pronounced work-hardening behavior observed in this sample. During plastic deformation of metals, the flow stress (σ) can be expressed as follows [55,56]:

$$\sigma=\sigma_0+\alpha M\mu b\sqrt{\rho_d} \quad (3)$$

where σ_0 is the flow stress required to move a dislocation in the lattice when $\rho\approx0$, α is a constant, M is the Talor factor, μ is the shear modulus, b is the magnitude of the Burgers vector of dislocation, and ρ_d is the dislocation density. The flow stress exhibits a positive correlation with dislocation density, explaining the origin of its high strength.

In D0 and D90, lower dislocation multiplication reveals another mechanism promotes plastic deformation. And it was identified as more dislocation dissociation products such as higher P_{sf} and HCP. Thus, the deformation process can be understood as a competition between dislocation multiplication and dissociation. It is evident that in D45, dislocation multiplication serves as the primary mechanism. Whereas in D90, the dissociation product, HCP, confirms dissociation dominate plastic deformation. Three samples exhibit different crystal orientation and grain boundary

density aligned with the loading direction. Further discussion is about that how the two factors influence the deformation mechanism.

It is acceptable that the influence of grain orientation on deformation can be descripted by Schmid factor. The Schmid factor influences not only the activation of perfect dislocation glide but also the dissociation of a perfect dislocation into leading and trailing partial dislocations (LPD and TPD). The tendency for dissociation can be quantitatively described by the effective stacking fault energy (γ_{eff}), which depends on the applied shear stress (τ_A). The relationship is given by:

$$\gamma_{\mathrm{eff}} = \gamma_0 + \frac{m_t - m_l}{2}\tau_A b \tag{4}$$

where γ_0 is the equilibrium stacking fault energy, m_t and m_l are the Schmid factors for the trailing and leading partial dislocations, respectively, and b is the magnitude of the Burgers vector. For grains oriented near <100> as predominantly found in D0, γ_{eff} decreases significantly under high stress to a value well below the critical threshold for dislocation dissociation [60], which stems from the negative value of $m_t - m_l$. This provides an explanation for the higher P_{sf} observed in D0. Specifically, grains oriented along <011> and <111> directions exhibit a higher γ_{eff}, which suppresses dislocation dissociation and consequently inhibits phase transformation [61].

Table 3 The Schmid factor for LPD (m_t) and TPD (m_l)[53]

Crystal direction	[111]	[110]	[100]
m_t	0.314	0.471	0.236
m_l	0.157	0.236	0.471

Grain size also affects the difficulty of dislocation dissociation by influencing the apparent stacking fault energy γ_{app}

$$\gamma_{app} = 2\rho_A(\Delta G_{ch}^{FCC\rightarrow HCP} + \Delta G_{ex}) + 2E_{int}^{FCC/HCP} \tag{5}$$

$$\Delta G_{ex} = 170.06exp(-D/18.55) \tag{6}$$

where ρ_A and $E_{int}^{FCC/HCP}$ are the density of atoms in a close-packed plane and the interfacial energy per unit area of the phase boundary between FCC and HCP phases, respectively. $\Delta G_{ch}^{FCC\rightarrow HCP}$ and ΔG_{ex} are the contributions of the chemical composition and grain size to Gibbs free energy difference, respectively. As shown in Eq (5) and (6) [62,63], other factors except grain size are the same among the three samples. A larger grain size reduces γ_{app} and thus lowers the barrier for dislocation dissociation.

D0 and D45 possess similar equivalent grain sizes (D0: 21.3 μm, D45: 21.9 μm measured by intercept lengths from Fig.S1e and d, respectively), yet they exhibit different rates of stacking fault growth, again indicating that dislocation dissociation occurs more readily in the <100> orientation. In contrast, compared with D0, D90 (50.9 μm measured by intercept lengths in Fig.S1c) has dominant orientations of <110> which are not considered favorable for dislocation dissociation. Nevertheless, the fractured D90 sample shows nearly an order of magnitude (≈ ten times) more HCP phase than D0. This comparison suggests that a coarse grain size lowers SF forming barriers more effectively than a favorable crystallographic orientation does. Also, the reduction in γ_{app} decreases the barrier for SIMT. As a result, the coarse grain size exerts a stronger influence on the activation of SIMT than the orientation effect. Therefore, grain size plays a more important role than crystallographic orientation in regulating low-SFE alloys through deformation-mechanism.

4 Conclusions

In this study, LPBF was used to fabricate FCC high-entropy alloy samples with columnar structures and special grain orientation, enabling investigation of anisotropic deformation mechanism in low-SFE alloy. The main findings are:

1) The LPBF process, combined with low SFE, resulted in unique microstructures containing cellular dislocations, stacking fault (SF) networks, and nano-scale HCP phases. These features enhance work hardening. Under the specific cutting strategy, the preferred crystal orientation was designed to vary significantly with the loading direction: D0 showed <100> and <311>, D45 exhibited <111>, and D90 displayed <110>.

2) Mechanical properties were strongly dependent on loading direction. Rapid dislocation multiplication in D45 enhance its flow stress, contributing to the highest strength (YS: 557 MPa, UTS: 788 MPa). D90 showed the highest elongation (64.8%) due to extensive SIMT.

3) The deformation mechanism exhibits a dependence on crystallographic orientation, as manifested in the relative contributions of SIMT and dislocation multiplication. Specifically, dislocation-mediated plasticity predominates in grains with a strong {111} texture. In contrast, the {100}-oriented grains facilitate the dissociation of perfect dislocations, generating a high density of stacking faults that act as precursors for twinning and subsequent SIMT.

4) Grain size variation exerts a more pronounced effect on SIMT than does a favorable

crystallographic orientation. Despite the <100> orientation favoring dislocation dissociation, D90 exhibits a higher fraction of deformation-induced HCP products. This anomaly is attributed to its larger grain size, which reduces the apparent stacking fault energy during deformation.

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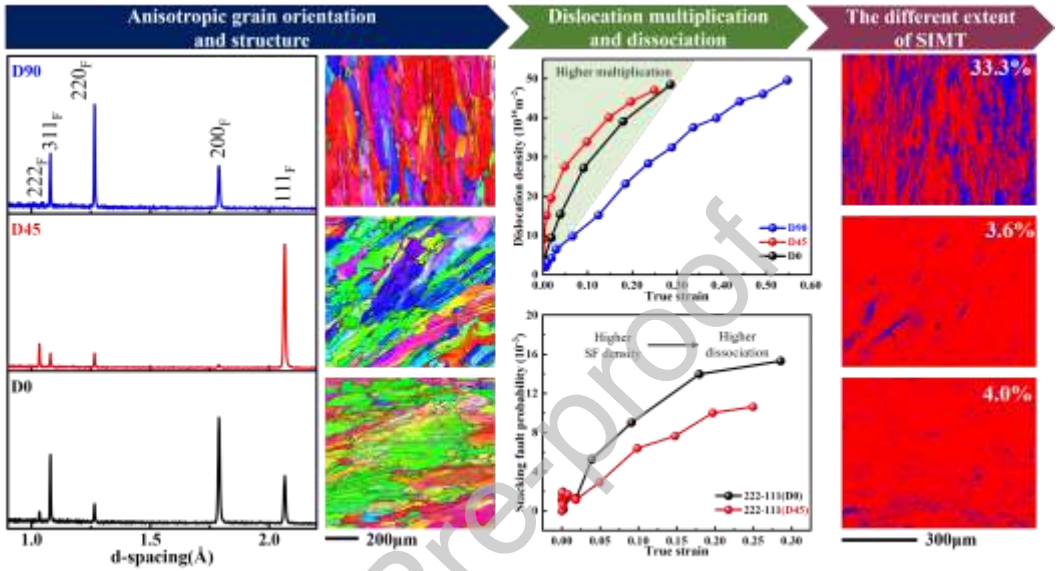
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